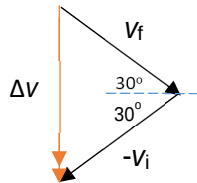


		u_x 0000000
4	C	Change in momentum = mass x change in velocity. Change in velocity = final velocity – initial velocity. However, velocity is a vector and vectors cannot be “simply” subtracted. Thus, the subtraction has to be changed into an addition: $\Delta v = v_f - v_i = v_f + (-v_i)$. From the vector addition diagram, since $v_f = v_i = u$ $\Delta v = 2u \sin 30^\circ = u$ $\Delta p = m \Delta u = mu$ 
5	A	When the truck is empty by Newton's second law